1. Document preprocessing and embedding creation



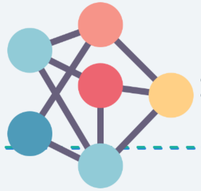
Collection of data gathered from the King Khalid Platform through web scraping and manually



Preprocess text into smaller chunks
e.g. groups of 10 sentences



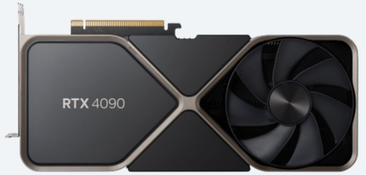
Smaller document chunks
(used as context for the LLM)



Embedding model turns text/query into
numerical representation
e.g. sentence-transformers,
many options here on Hugging Face



Store embeddings in
torch.tensor
(fast for even ~100k+ embeddings)



All happens on a local GPU



Optional: Use a vector database/index
if your dataset is really large.



Student



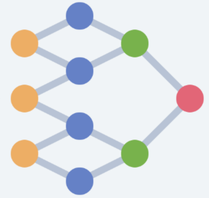
Ask a query

What are admission policies and available majors?

What are admission policies and available majors?

\+

Query + relevant passages



LLM generates output based on
context related to query

Generated text and resources



AIKKU



AIKKu Chat Web APP Interface

2. Search and answer